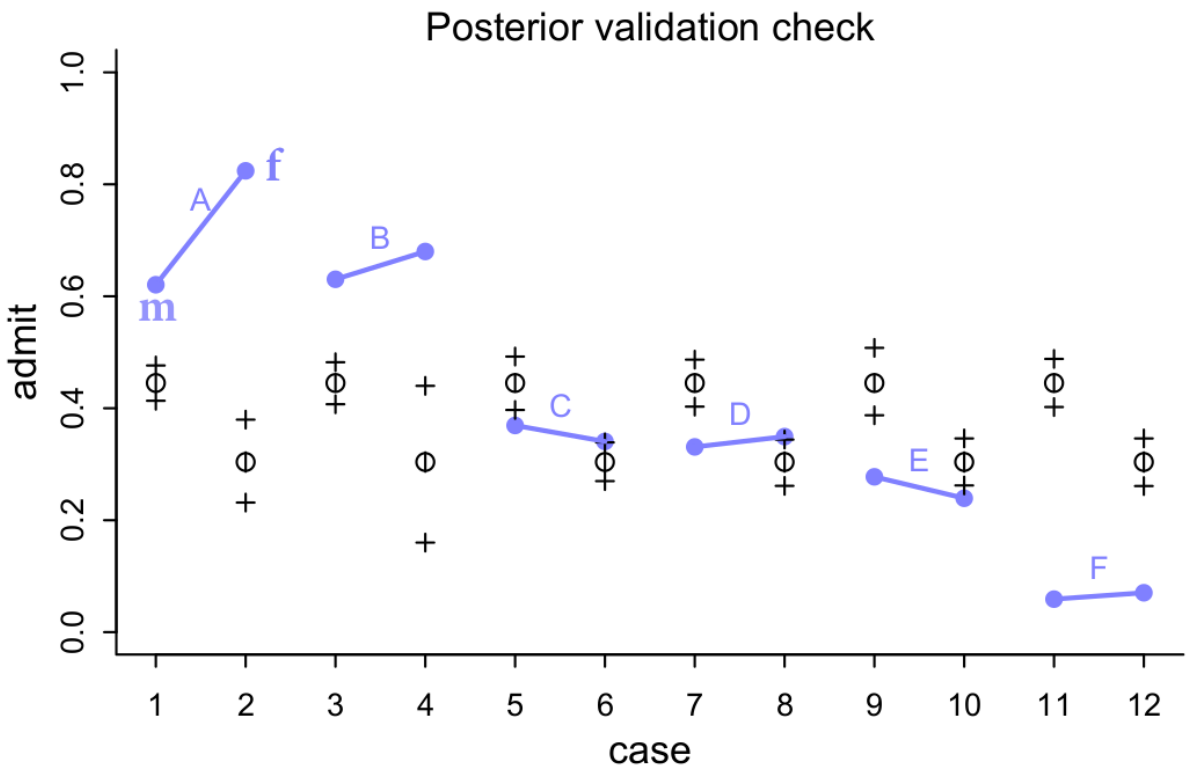
GLMs in practice

GLMs in practice on examples II

Conclusions of the UB Berkeley Admission data analysis



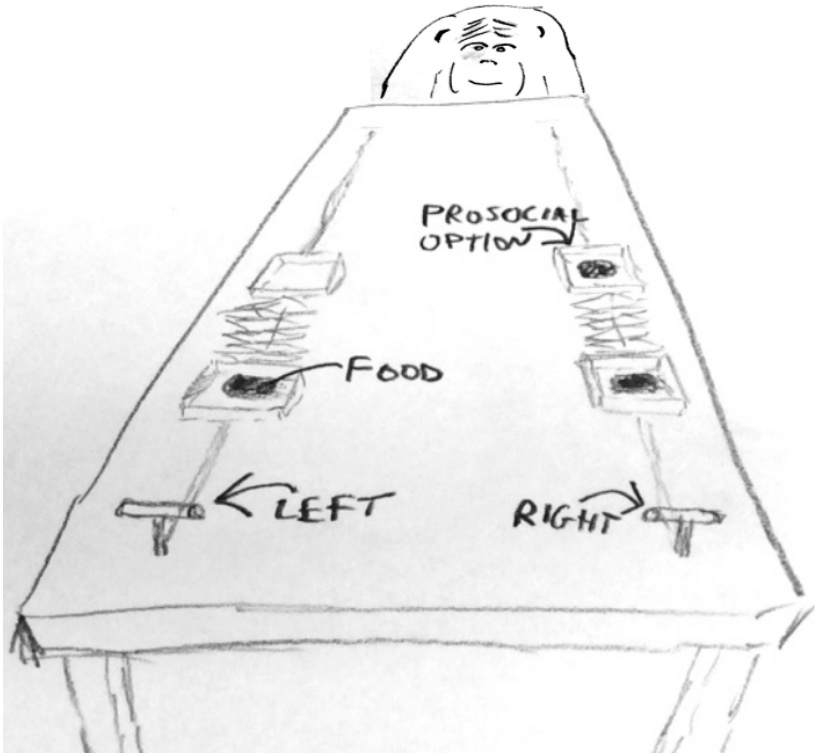
(Figure 11.5, SR book)

Female admitted more in all but two departments.

Conclusion of the UC Berkeley Admissions data analysis

- Females apply more to most selective departments $\rightarrow$ overall rate of admission lower (see the proportions of M/F applications per department on p344, SR book.
- Careful about causal interpretation.
- No evidence for a clear $G \rightarrow A$ effect. Lots of evidence for a $G \rightarrow D \rightarrow A$ indirect effect.
- Total causal influence of G(ender) is still strong. Results in disenfranchisement.
- But effective intervention very different.

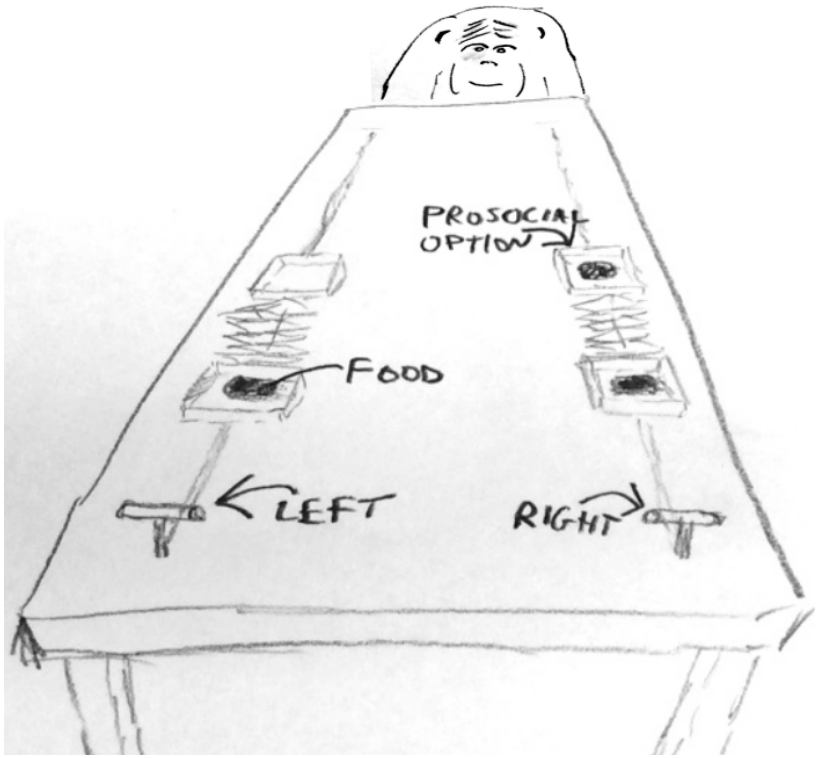
Prosocial chimpanzees experiment





Prosocial chimpanzees experiment

- Two conditions: (1) partner or (2) alone
- Two options: (1) prosocial or (2) asocial
- Two outcomes: (1) left lever or (2) right lever



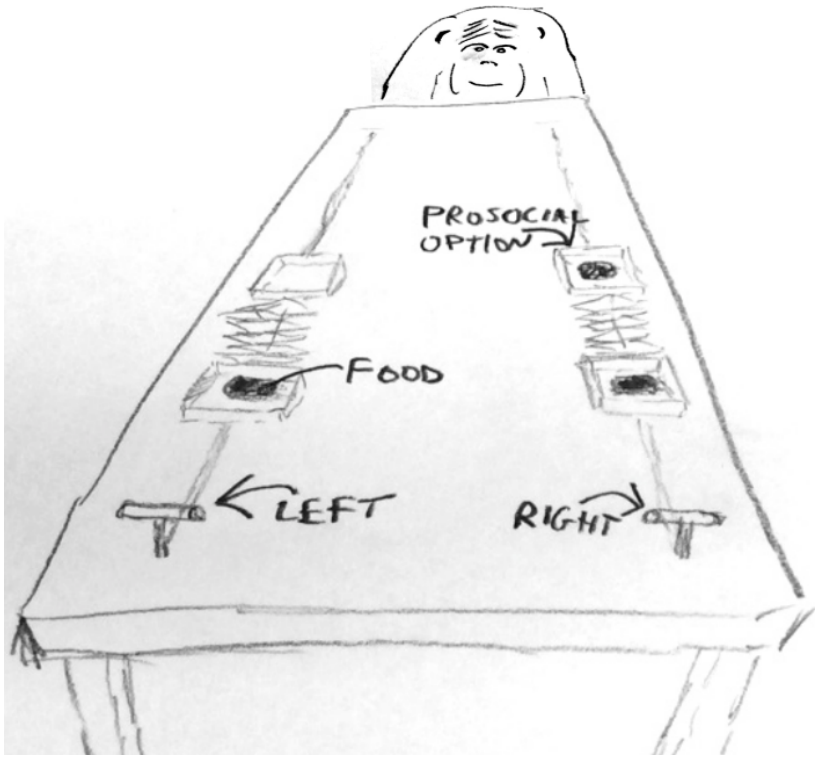
Prosocial chimpanzees experiment

```
data("chimpanzees")
head(chimpanzees)
```

```
##   actor recipient condition block trial prosoc_left chose_prosoc
## 1     1        NA         0     1     2           0           1
## 2     1        NA         0     1     4           0           0
## 3     1        NA         0     1     6           1           0
## 4     1        NA         0     1     8           0           1
## 5     1        NA         0     1    10           1           1
## 6     1        NA         0     1    12           1           1
```

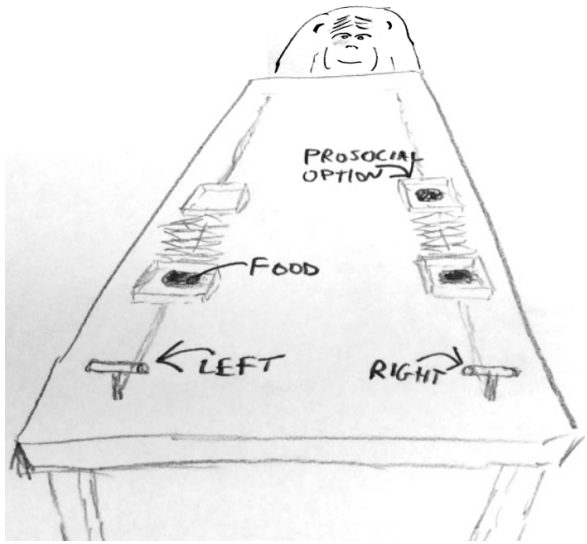
Prosocial chimpanzees experiment

- Two conditions: (1) partner or (2) alone
- Two options: (1) prosocial or (2) asocial
- Two outcomes: (1) left lever or (2) right lever
- Want to predict the outcome as a function of condition and which side the option is one
  - Do chimpanzees prefer the left lever when partner present a prosocial option on the left? $\rightarrow$ interaction!



Prosocial chimpanzees experiment

- Coding treatments:
  - right/no-partner: (1)
  - left-no-partner: (2)
  - right/partner: (3)
  - left/partner: (4)



$$
\begin{aligned}
L_i &\sim \text{Binomial}(1, p_i) \\
\text{logit}(p_i) &= \alpha_{ACTOR[i]} + \beta_{TREATMENT[i]} \\
\alpha_j &\sim \text{to be determined} \\
\beta_k &\sim \text{to be determined}
\end{aligned}
$$

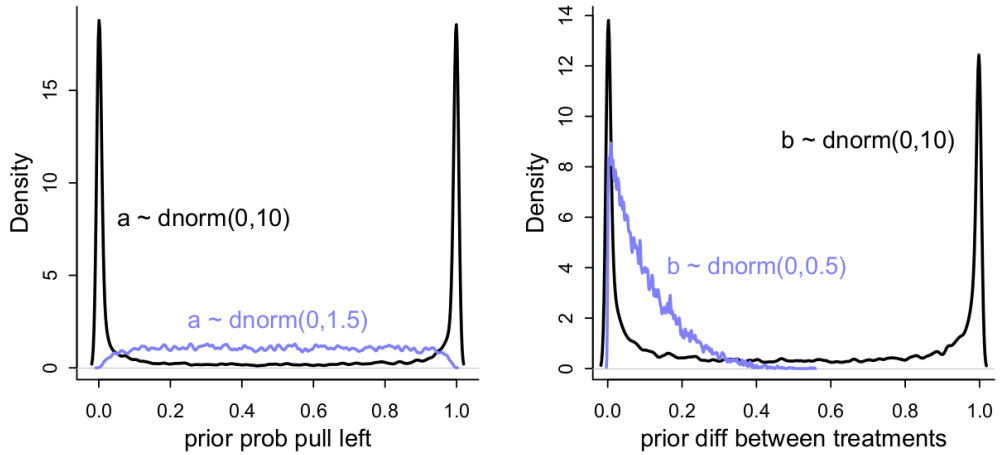
$\text{Binomial}(1, p)$ often called logistic regression; same as $\text{Bernoulli}(p)$.

Logit link priors

- Link functions: link parameters of distribution to linear model
- Pick the distribution of the outcome, link function matches
- $\text{logit}(p_i) = \log \frac{p_i}{(1-p_i)}$ (the log-odds) and the inverse-logit is the logistic: $p_i = \frac{\exp(\alpha+\beta x_i)}{1+\exp(\alpha+\beta x_i)}$

Logit link priors

- Prior on logit scale not same shape as prior on probability scale
- Weakly informative priors on these scales seem very restrictive. Very small SD (see pg 327-329).



- Also see the practice problem at the end of Chapter 11 to compare the fit with `quap()` to the fit with `ulam()`.

Chimpanzees statistical model output

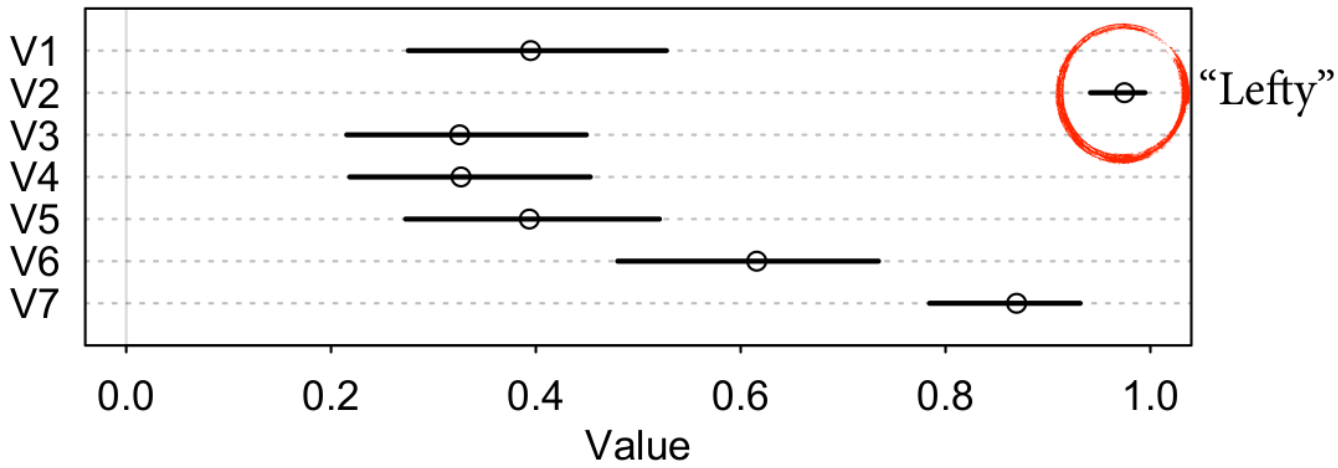
```
# particles in 11-dimensional space
m11.4 <- ulam(
    alist(
        pulled_left ~ dbinom( 1 , p ) ,
        logit(p) <- a[actor] + b[treatment] ,
        a[actor] ~ dnorm( 0 , 1.5 ),
        b[treatment] ~ dnorm( 0 , 0.5 )
    ) ,
    data=dat_list , chains=4 )
precis( m11.4 , depth=2 )
```

		mean	sd	5.5%	94.5%	n_eff	Rhat
Chimpanzees	a[1]	-0.44	0.34	-0.97	0.11	736	1
	a[2]	3.90	0.77	2.78	5.22	921	1
	a[3]	-0.75	0.34	-1.29	-0.20	886	1
	a[4]	-0.74	0.34	-1.28	-0.19	770	1
	a[5]	-0.44	0.34	-0.98	0.08	832	1
	a[6]	0.48	0.34	-0.08	1.02	854	1
	a[7]	1.96	0.41	1.29	2.61	847	1
Treatments: RN	b[1]	-0.05	0.29	-0.51	0.42	781	1
LN	b[2]	0.48	0.29	0.03	0.94	657	1
RP	b[3]	-0.39	0.28	-0.83	0.07	669	1
LP	b[4]	0.36	0.29	-0.11	0.81	732	1

Chimpanzees statistical model individual differences

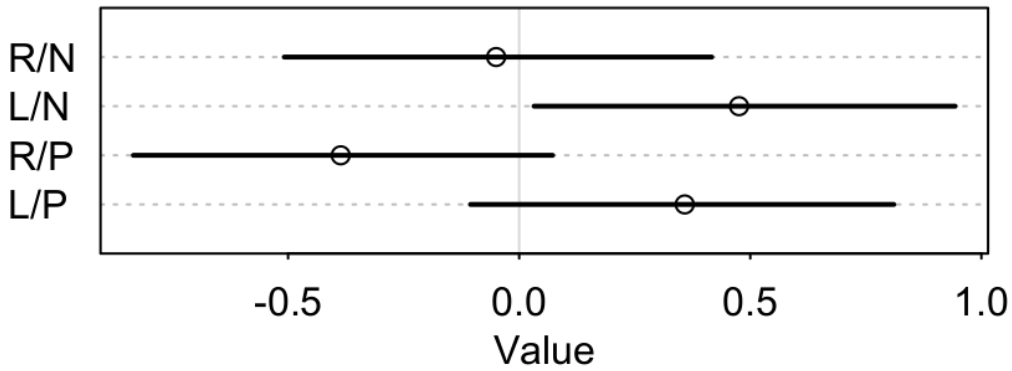
R code 11.11

```
post <- extract.samples(m11.4)
p_left <- inv_logit( post$a )
plot( precis( as.data.frame(p_left) ) , xlim=c(0,1) )
```



Chimpanzees statistical model treatment differences

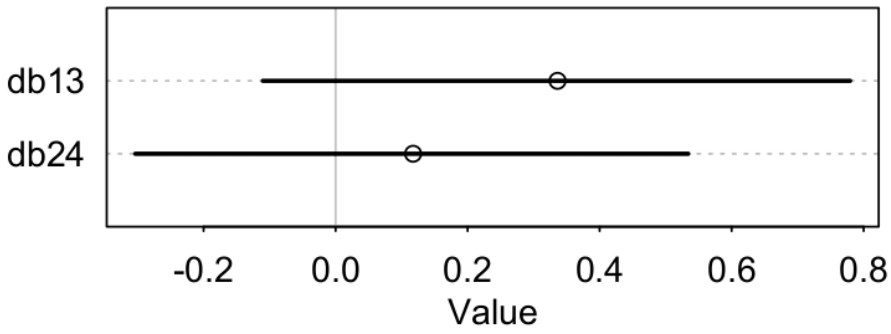
```
labs <- c("R/N","L/N","R/P","L/P")
plot( precis( m11.4 , depth=2 , pars="b" ) , labels=labs )
```



Chimpanzees statistical model treatment differences

Need to look at contrasts between partner/no partner treatments

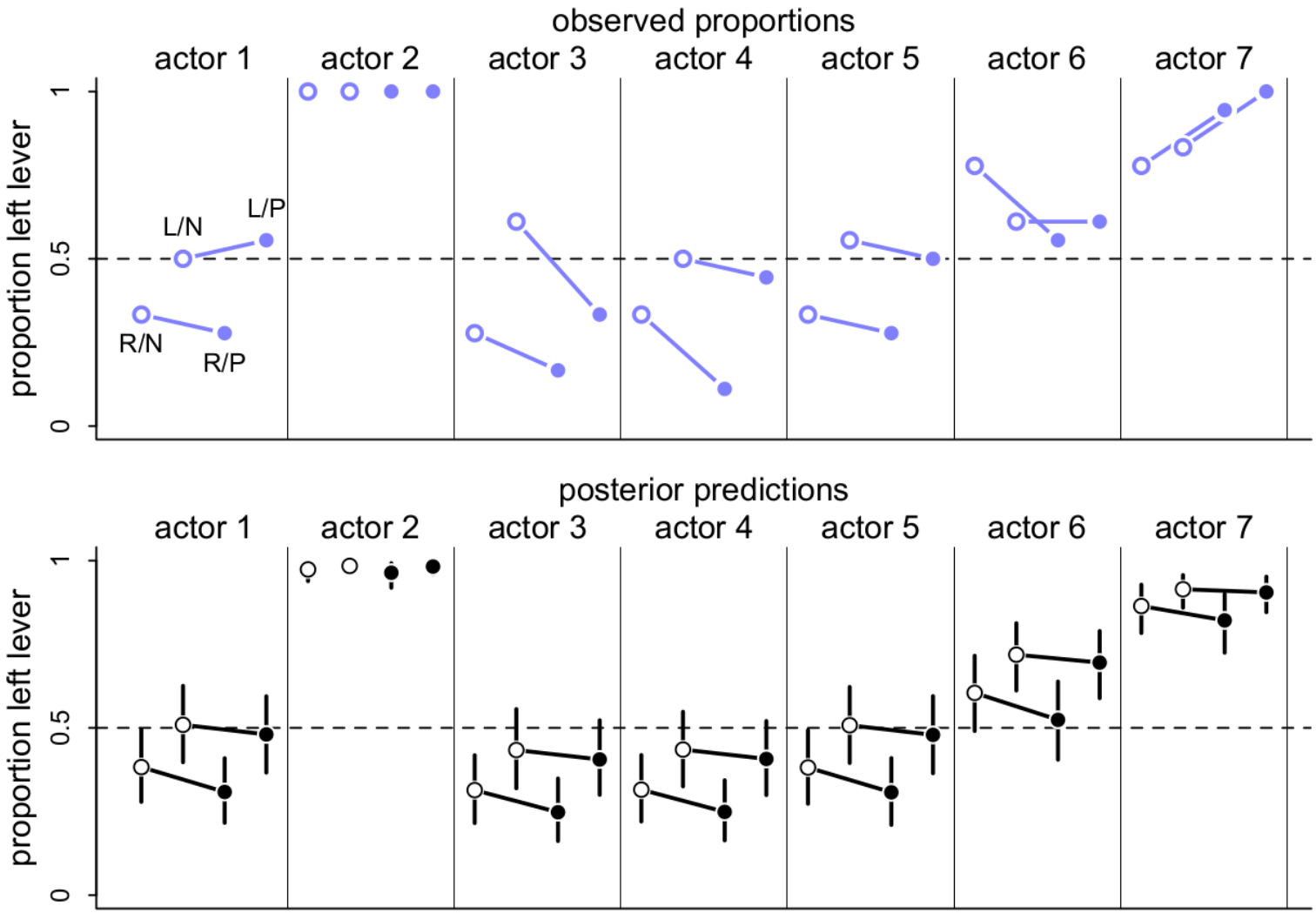
```
diffs <- list(db13  = post$b[,1]-post$b[,3],
   db24  = post$b[,2]-post$b[,4])
plot(precis(diffs))
```



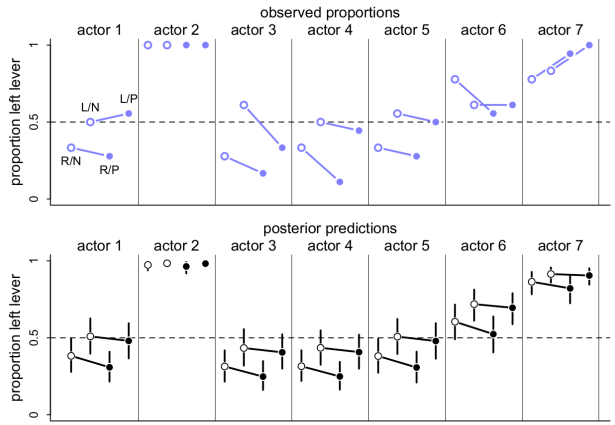
scale is log-odds of pulling the left lever)

Overall, no compelling evidence of prosocial choice in this experiment.

Chimpanzees statistical model: Observed proportions & Posterior predictions



Chimpanzees statistical model: Observed proportions & Posterior predictions



- Model expects almost no change when adding a partner.
- Most variation in predictions comes from the actor intercept $\rightarrow$ Handedness seems to be a big story in this experiment.
- Data themselves show additional variation; some actors possibly respond more to the treatment than others do. Might consider a model, with unique treatment parameters. We'll leave such possibility when we introduce multilevel models.

Comparing the model to one without interactions

Only for the sake of the exercise; the experiment drove us to choose `m11.4`

```
m11.5 <- ulam(
    alist(
        pulled_left ~ dbinom( 1 , p ) ,
        logit(p) <- a[actor] + bs[side] + bc[cond] ,
        a[actor] ~ dnorm( 0 , 1.5 ),
        bs[side] ~ dnorm( 0 , 0.5 ),
        bc[cond] ~ dnorm( 0 , 0.5 )
    ) ,
    data=dat_list2 , chains=4 , log_lik=TRUE )
```

R code 11.19

```
compare( m11.5 , m11.4 , func=LOO )
```

```
        LOO pLOO dLOO weight    SE  dSE
m11.5 531.2  7.9  0.0   0.66 19.17   NA
m11.4 532.6  8.7  1.4   0.34 19.01 1.28
```

OK model without interaction was sufficient as suggested by posterior predictive checks. Inspect `m11.5` posterior distributions!

Absolute and relative effects

- Shark vs deer...
- Deer - 100 people/year
- Vending machines - 13 people/year
- Selfies - 12 people/year
- Falling out of bed - 450 people/year
- Deadliest animal? Mosquitoes - 1 million people/year!!
- Sharks 10 people/year

Absolute and relative effects

- Posterior prediction, looking at the outcome focuses in the *absolute effect*: the difference a counter-factual change in a variable might make on an absolute scale of measurement (here probability of an event).
- Looking at the parameters (common in the result analysis of GLMs) focuses on *relative effects*, that are proportional changes in the odds of an outcome in logistic regression. Not enough to tell us whether a variable is important or not.
- Both absolute and relative effects are important. The latter can be important to make causal inference when conditioning is necessary; when other baselines vary. Conditional on specific circumstances, we need relative risks. For general predictions ("out-of-context"), absolute risks often make more sense.

Risk communication

- Many people mistake relative risk for absolute risk
- Example
  - 1/1000 women develop blood clots
  - 3/1000 women on birth control develop blood clots
  - $\rightarrow$ 200% increase in blood clots. But the change in probability is only 0.002; pregnancy is much more dangerous than blood clots.

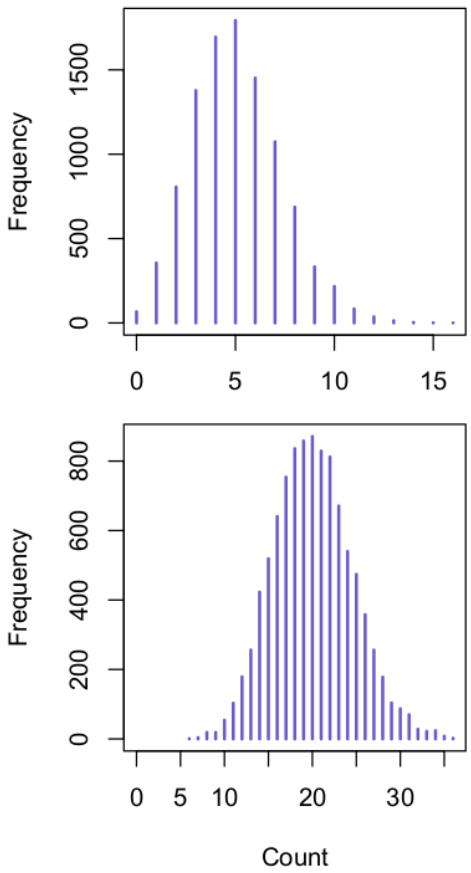


Relative effects may exagerate the importance of predictor(s): good to scare people or being published. Not so good for public health or scientific progress. But needed for causal inference.

Poisson GLMs

Poisson Distribution

$$\begin{aligned} y &\sim \text{Poisson}(\lambda) \\ E[y] &= \lambda \\ \text{var}(y) &= \lambda \end{aligned}$$



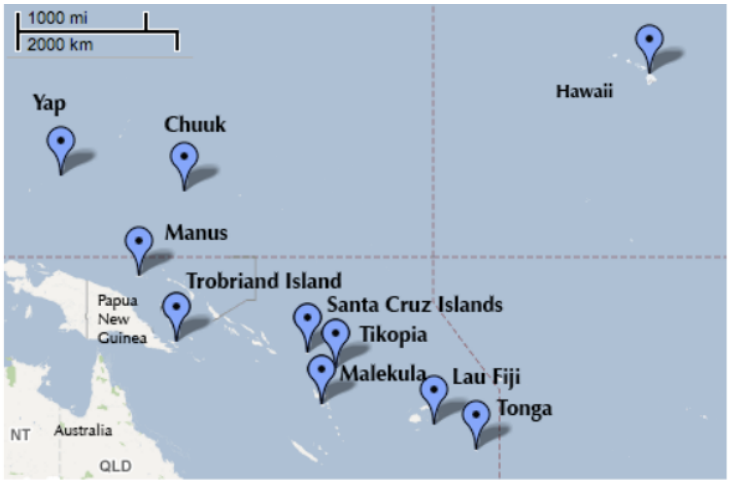
- Counts without upper limit; constant expected value
- Single parameter: events per unit of time/distance
- Variance equal to the mean

Examples: soccer goals, fission events, photons striking a detector, DNA

Poisson regression data example

Oceanic tool complexity

culture	population	contact	total_tools	mean_TU
Malekula	1100	low	13	3.2
Tikopia	1500	low	22	4.7
Santa Cruz	3600	low	24	4.0
Yap	4791	high	43	5.0
Lau Fiji	7400	high	33	5.0
Trobriand	8000	high	19	4.0
Chuuk	9200	high	40	3.8
Manus	13000	low	28	6.6
Tonga	17500	high	55	5.4
Hawaii	275000	low	71	6.6



- Complexity of toolkit proportional to the magnitude of the population?
- Contact with other islands moderates the impact?

Oceanic tool complexity

Kline & Boyd 2010 Population size predicts technological complexity in Oceania

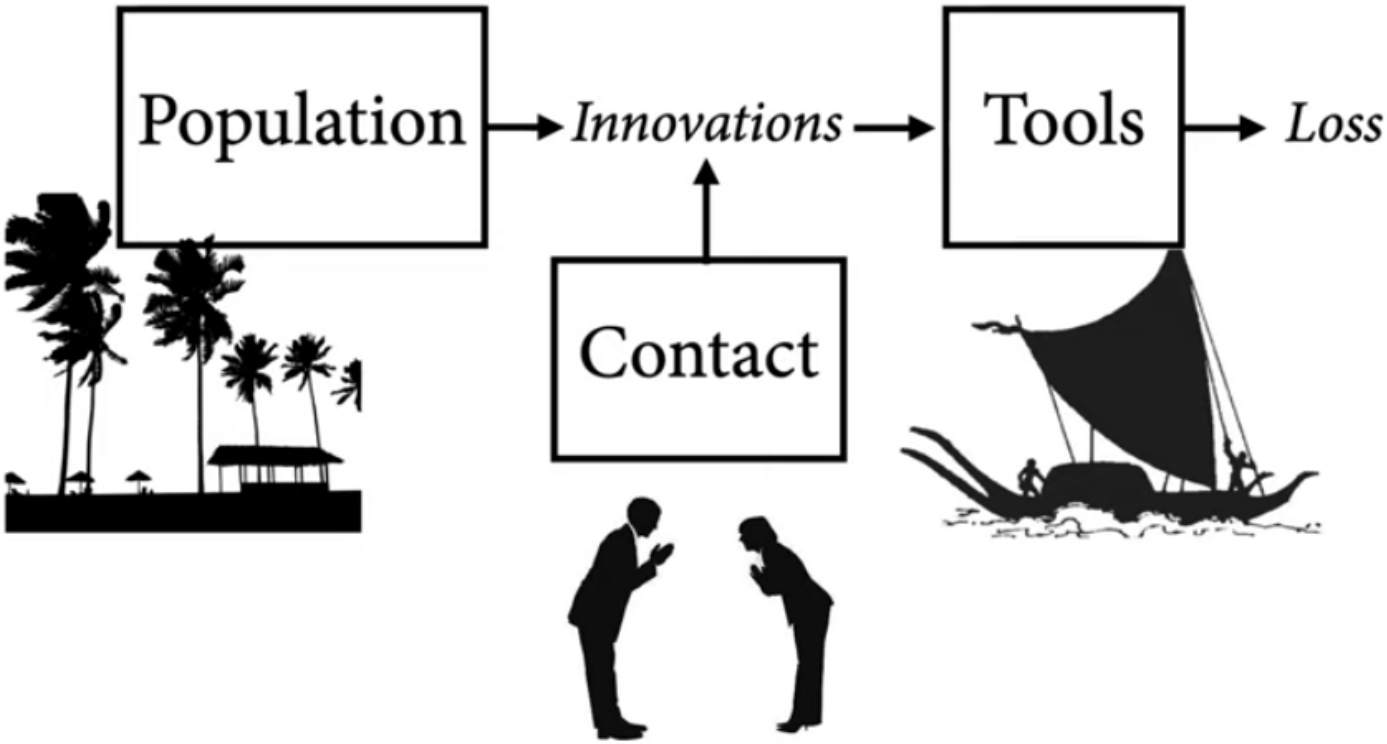
Influence of **population size** and **contact** on **total tools**

```
data("Kline")

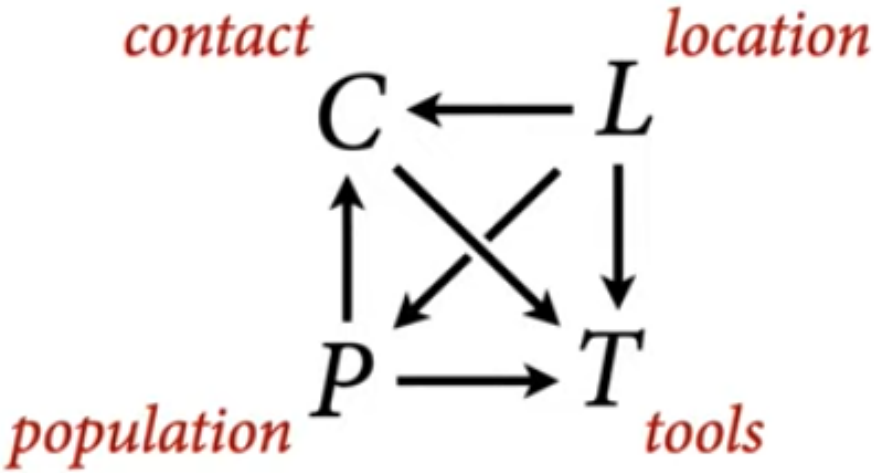
Kline
```

```
##         culture population contact total_tools mean_TU
## 1     Malekula       1100     low          13     3.2
## 2      Tikopia       1500     low          22     4.7
## 3   Santa Cruz       3600     low          24     4.0
## 4          Yap       4791    high          43     5.0
## 5     Lau Fiji       7400    high          33     5.0
## 6    Trobriand       8000    high          19     4.0
## 7        Chuuk       9200    high          40     3.8
```

Oceanic tool complexity

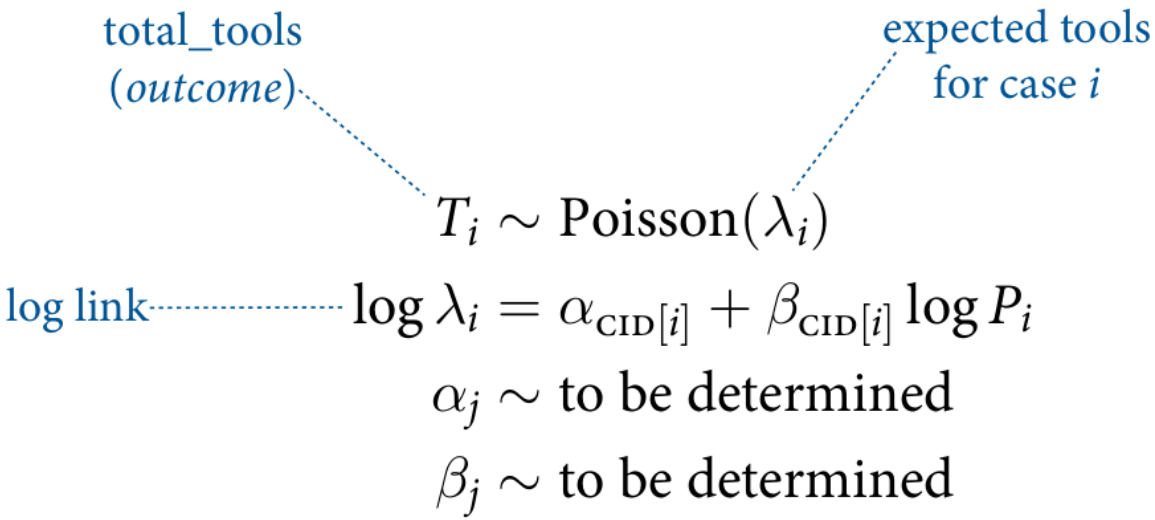


Oceanic tool complexity



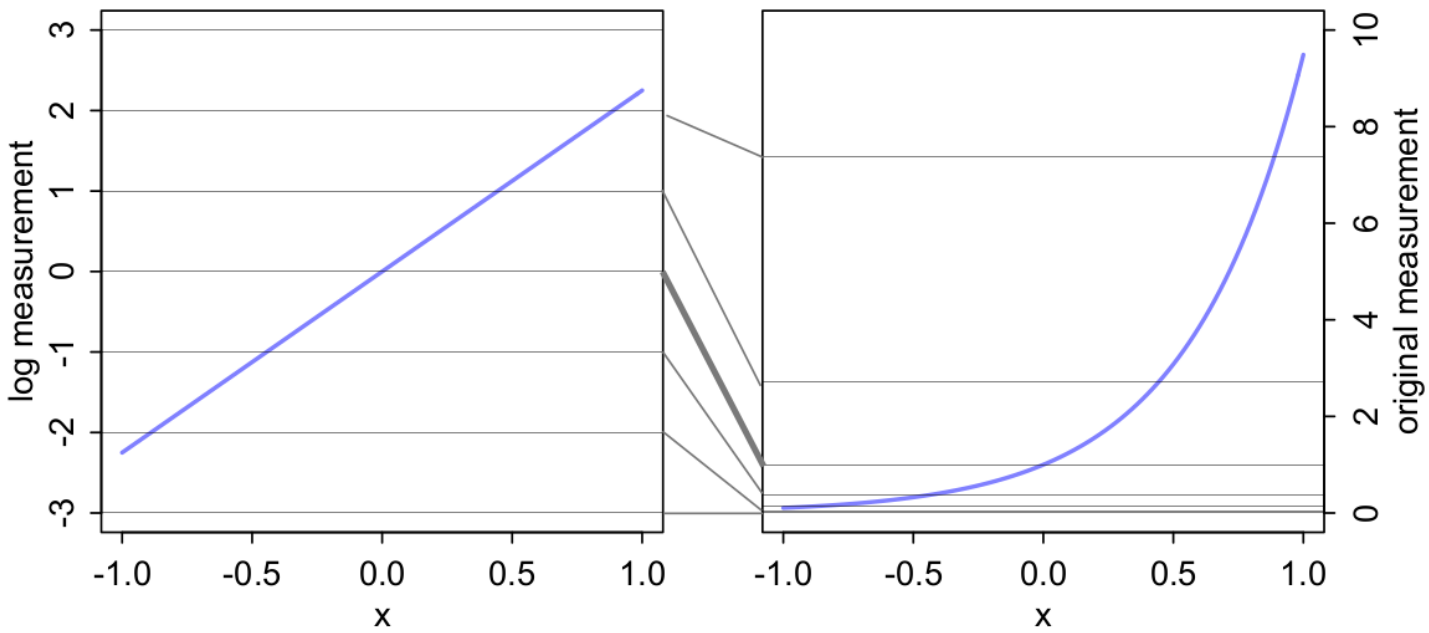
Anatomy of a Poisson GLM

Oceanic tool complexity



Log link for the Poisson regression

- Goal: map the linear combination space to positive real values
- All negative numbers mapped to [0,1]
- All positive numbers mapped to $[1,\infty)$



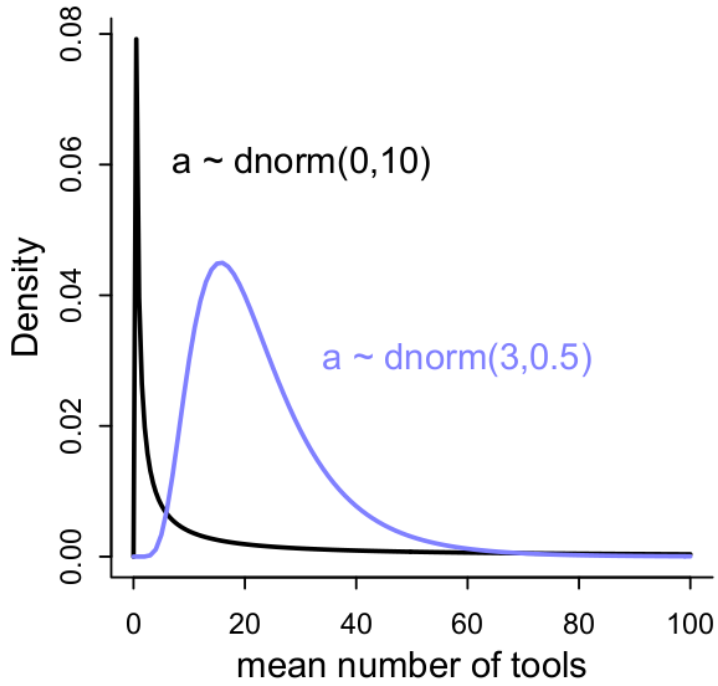
Priors & log link

- Log link is not intuitive $\rightarrow$ simulate

$$
\begin{aligned}
T_i &\sim \text{Poisson}(\lambda_i) \\
\log \lambda_i &= \alpha \\
\alpha &\sim \text{Normal}(0, 10)
\end{aligned}
$$

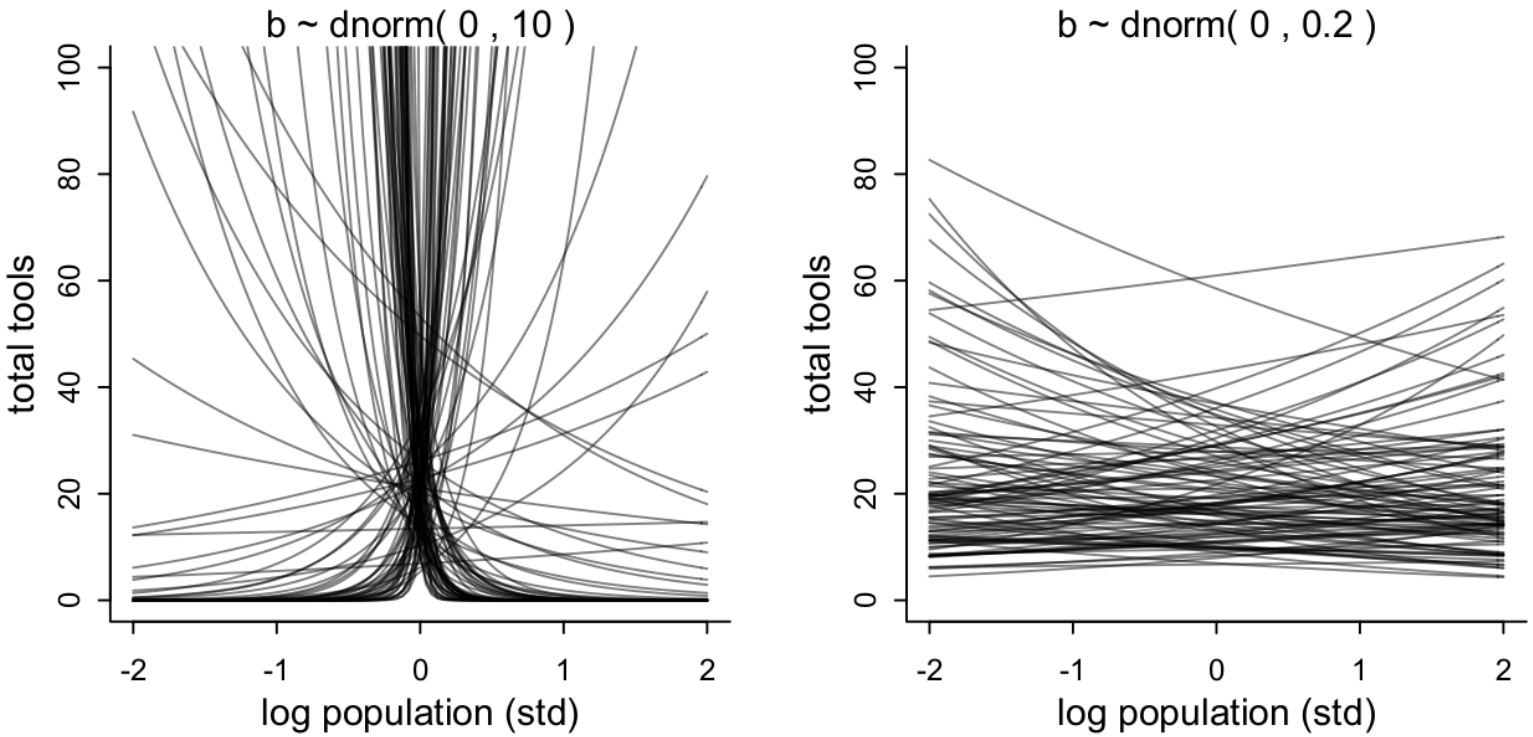
```
a <- rnorm(1e4,0,10)
lambda <- exp(a)
mean( lambda )
```

```
[1] 9.622994e+12
```



Priors & log link

- Slopes equally unintuitive



Poisson GLM in practice

Oceanic tool complexity: two models

```
dat <- list(
    T = d$total_tools ,
    P = d$P ,
    cid = d$contact_id )

# intercept only
m11.9 <- ulam(
    alist(
        T ~ dpois( lambda ),
        log(lambda) <- a,
        a ~ dnorm(3,0.5)
    ), data=dat , chains=4 , log_lik=TRUE )


# interaction model
m11.10 <- ulam(
    alist(
        T ~ dpois( lambda ),
        log(lambda) <- a[cid] + b[cid]*P,
        a[cid] ~ dnorm( 3 , 0.5 ),
        b[cid] ~ dnorm( 0 , 0.2 )
    ), data=dat , chains=4 , log_lik=TRUE )
```

Oceanic tool complexity: two models

```
compare( m11.9 , m11.10 , func=LOO )
```

R code 11.49

```
          LOO pLOO dLOO weight    SE   dSE
m11.10  85.5  7.1  0.0      1 13.22    NA
m11.9  141.1  8.0 55.5      0 33.33 32.78
Warning messages:
1: Some Pareto k diagnostic values are too high. See help('pareto-k-diagnostic') for details.
```

- Warning indicates strongly influential points.
- Look at those pLOO values
- How can a model with more parameters (m11.10) have fewer effective parameters?
- No relationship to parameter count; not unusual and not a mistake - have to give up on intuition

Oceanic tool complexity: two models

```
compare( m11.9 , m11.10 , func=LOO )
```

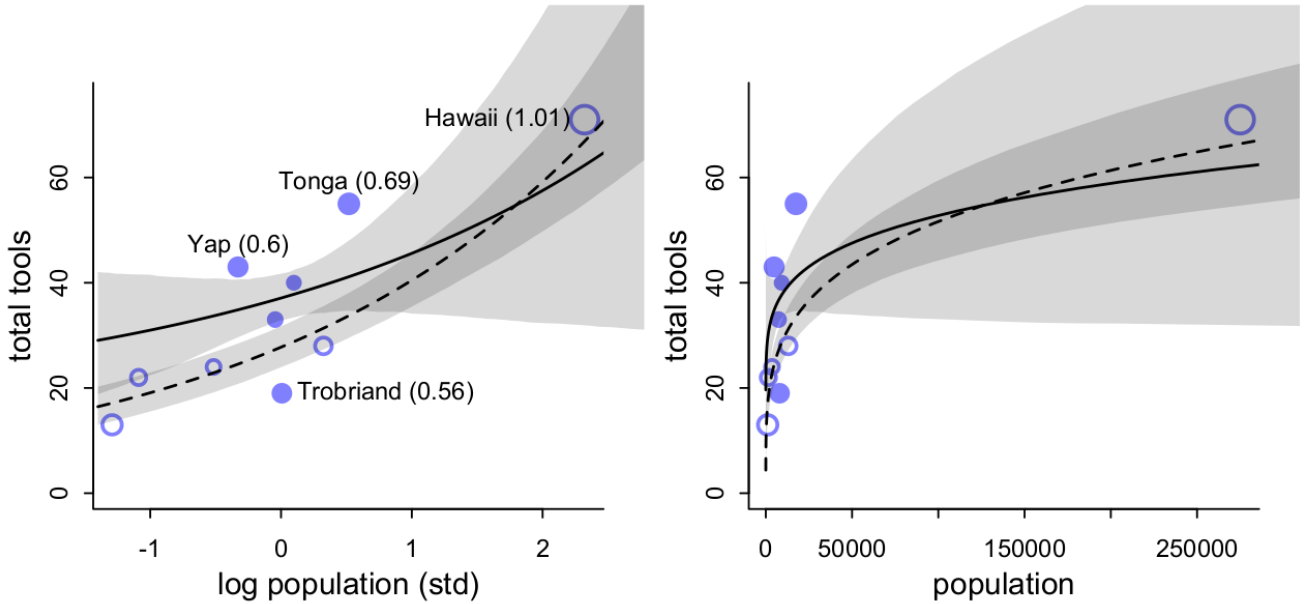
R code 11.49

```
          LOO pLOO dLOO weight    SE   dSE
m11.10   85.5  7.1  0.0      1 13.22    NA
m11.9   141.1  8.0 55.5      0 33.33 32.78
Warning messages:
1: Some Pareto k diagnostic values are too high. See help('pareto-k-diagnostic') for details.
```

- How can a model with more parameters (m11.10) have fewer effective parameters?
- No relationship to parameter count; not unusual and not a mistake - have to give up on intuition
- Penalty really measures how much of a difference it makes in posterior when we drop an society
- It isn't the number of parameters that matter – it is how they are structured and how they relate

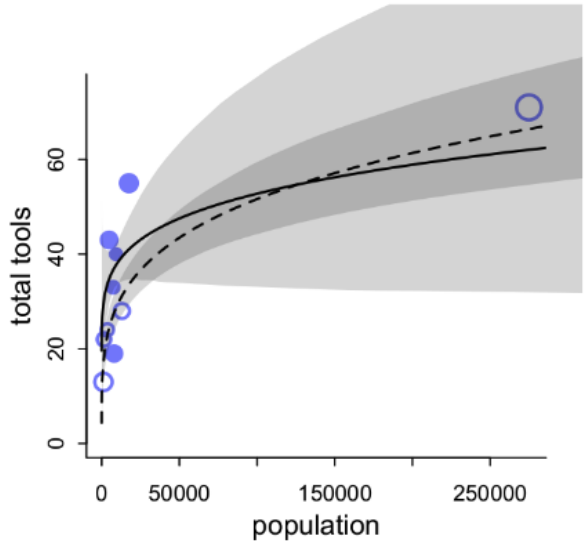
Oceanic tool complexity: Hawaii has leverage

Posterior Predictions



- Point size proportional to Pareto-k diagnostic value
- Solid points high contact societies and open points low contact – historically

Model discussion



What do you notice? Describe the plot.

Model discussion

- Hawaii has an extreme population size and the most tools.
- Trends cross: high contact islands level off sooner but why would low contact islands eventual pass them?
- Trend for society with high contacts (solid) is higher than that for society with low contacts (dashed) when population size is low.
- But when the model allows it to be smaller :-/
- Prediction windows very large.
- The model is actually saying it has no idea where the trend for high contact/high population islands goes because...there is no such island in our data. Only Hawaii and it has a low contact.
- See the scientific model on p355-356 in the SR book.